

Assignment 7 - Sequential Agentic Workflows using LangGraph

Objective

The objective of this assignment is to develop sequential agentic workflows using LangGraph and Gemini 2.5 Flash. The workflows demonstrate multi-step execution, state management, tool calling, and LLM-based decision making.

Technologies Used

- Python
 - LangGraph
 - Gemini 2.5 Flash
 - Google GenAI SDK
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USE CASE 1: Telecom SIM Activation & Fraud Verification Workflow

Business Problem

A telecom company wants to automate SIM activation while ensuring customer verification and fraud-risk analysis.

Workflow Steps

Step 1 - KYC Verification Tool

A tool was developed to validate:

- Aadhaar status
- PAN status
- Document validity

Possible outputs:

- VERIFIED
 - INVALID_DOCUMENT
 - PENDING_VERIFICATION
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Step 2 - Gemini Fraud Detection

Gemini 2.5 Flash analyzes:

- Customer location
- Number of previous SIM requests
- KYC verification status

Gemini classifies fraud risk as:

- LOW_RISK
 - MEDIUM_RISK
 - HIGH_RISK
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Step 3 - Final Activation Decision

Based on fraud classification:

- LOW_RISK → SIM Activated
 - MEDIUM_RISK → Manual Review
 - HIGH_RISK → Reject Application
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Telecom Workflow Result

The workflow successfully performs:

- Sequential execution
 - State passing between nodes
 - Gemini-based fraud analysis
 - Automated activation decision
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USE CASE 2: Healthcare Appointment Prioritization Workflow

Business Problem

A hospital wants to automate appointment prioritization based on patient symptoms and urgency.

Workflow Steps

Step 1 - Symptom Severity Tool

The tool analyzes:

- Fever
- Oxygen level
- Heart rate
- Symptom duration

Possible outputs:

- STABLE
 - MODERATE
 - CRITICAL
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Step 2 - Gemini Medical Prioritization

Gemini 2.5 Flash evaluates:

- Severity level
- Patient age
- Oxygen level
- Heart rate

Gemini classifies:

- EMERGENCY
 - PRIORITY_CONSULTATION
 - REGULAR_CONSULTATION
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Step 3 - Final Consultation Assignment

Based on priority:

- EMERGENCY → ICU / Emergency Ward
 - PRIORITY_CONSULTATION → Specialist Doctor
 - REGULAR_CONSULTATION → General Physician
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Healthcare Workflow Result

The workflow successfully performs:

- Sequential medical prioritization
 - LLM-based reasoning
 - Automated consultation assignment
 - State management across workflow stages
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LangGraph Concepts Used

State Management

State objects were used to pass information between workflow nodes.

Sequential Workflow Execution

Each node depends on outputs from previous nodes, ensuring sequential execution.

Tool Calling

Custom Python functions were used as workflow tools.

Gemini-based Decision Intelligence

Gemini 2.5 Flash was integrated for reasoning and classification tasks.

Conclusion

The assignment successfully demonstrates enterprise-style sequential agentic workflows using LangGraph and Gemini 2.5 Flash. Both telecom and healthcare workflows showcase state management, sequential execution, tool integration, and LLM-powered decision making.

Output Examples

Telecom Workflow Output

KYC STATUS: VERIFIED
FRAUD RISK: LOW_RISK
FINAL DECISION: SIM Activated

Healthcare Workflow Output

SEVERITY STATUS: CRITICAL

CONSULTATION PRIORITY: EMERGENCY

FINAL ASSIGNMENT: ICU / Emergency Ward